

Root relationships- Quadratics

Basic skills

Q1. a) $3x^2 + 6x - 9 = 0$ could $\div 3$
 $(3x - 3)(x + 3) = 0 \Rightarrow x = 1 \text{ or } -3$

b) $2x^2 - 10x + 6 = 0 \Rightarrow x^2 - 5x + 3 = 0$
cannot factorise so
$$x = \frac{5 \pm \sqrt{25 - 12}}{2} = \frac{5 \pm \sqrt{13}}{2}$$

Q2. a) $2x^2 + 8x - 10 = 0$
 $(2x - 2)(x + 5) = 0$
 $x = 1 \text{ or } -5$

so $\alpha = 1, \beta = -5$

b) $\alpha + \beta = 1 + -5 = -4 = -\frac{8}{2} \checkmark$

$\alpha\beta = 1 \times -5 = -5 = -\frac{10}{2} \checkmark$

c) $(x - 1)(x + 5) = 0$
 $(x - \alpha)(x - \beta) = 0$

$$\begin{aligned}
 d) \quad & (x - (\alpha + 1))(x - (\beta + 1)) = 0 \\
 & (x - 2)(x - -4) = 0 \\
 & (x - 2)(x + 4) = 0 \\
 & x^2 + 2x - 8 = 0
 \end{aligned}$$

Competency skills

Q3. $3x^2 + 14x - 3 = 0$

a) $\alpha + \beta = -\frac{14}{3}$, $\alpha\beta = \frac{-3}{3} = -1$

b) so $\beta = -\frac{1}{\alpha} \Rightarrow \alpha - \frac{1}{\alpha} = -\frac{14}{3}$

$$\Rightarrow \alpha^2 - 1 = -\frac{14}{3}\alpha \Rightarrow \alpha^2 + \frac{14}{3}\alpha - 1 = 0$$

$$3\alpha^2 + 14\alpha - 3 = 0 \quad \leftarrow \text{same eq.}$$

$$\alpha = \frac{-14 \pm \sqrt{14^2 + 36}}{6} \quad \leftarrow \text{note one is } \alpha, \text{ the other is } \beta$$

The point of the

$\alpha + \beta$, $\alpha\beta$ are not to find roots, but relate equations

c) new roots: $2\alpha, 2\beta$

So sum: $2\alpha + 2\beta = 2(\alpha + \beta) = 2 \times -\frac{14}{3}$

Product: $2\alpha \times 2\beta = 4\alpha\beta = -\frac{28}{3}$
 $= 4 \times -1 = -4$

So new eq has $-\frac{b}{a} = -\frac{28}{3}$, $\frac{c}{a} = -4$

Freedom to choose a

let $a = 3$ then

$b = 28$, $c = -12$

So $3x^2 + 28x - 12 = 0$

Q4. $2z^2 + 5z + 10 = 0$ α, β roots

a) $\alpha + \beta = -\frac{b}{a} = -\frac{5}{2}$, $\alpha\beta = \frac{c}{a} = \frac{10}{2} = 5$

new roots $\alpha+1, \beta+1$

Sum: $(\alpha+1) + (\beta+1) = \alpha + \beta + 2 = -\frac{5}{2} + 2 = -\frac{1}{2}$

Product: $(\alpha+1)(\beta+1) = \alpha\beta + \alpha + \beta + 1$
 $= 5 + -\frac{5}{2} + 1$
 $= \frac{7}{2}$

so for new equation

$$-\frac{b}{a} = -\frac{1}{2}, \quad \frac{c}{a} = \frac{7}{2}$$

let $a=2$, $b=-1$, $c=7$

Freedom
to
Pick a

so $2z^2 - z + 7 = 0$

b) $z \rightarrow z-1$ gives

$$2(z-1)^2 + 5(z-1) + 10 = 0$$

$$2(z^2 - 2z + 1) + 5z + 5 = 0$$

$$2z^2 + 1z + 7 = 0$$

$$2z^2 - z + 7 = 0$$

same as
before

works as

$$\begin{aligned} &(z - (\alpha+1)) \\ &(\underbrace{z-1} - \alpha) \\ &z \rightarrow z-1 \end{aligned}$$

commonly worked out by
 $z - (\alpha+1) = 0 \quad \alpha = z-1$
so $z \rightarrow z-1$

c) could use either method, but as symmetric substitution is 'easier'

For $2\alpha+1, 2\beta+1$

Need $z - (2\alpha+1) = 0$ so $\alpha = \frac{z-1}{2}$

So need $z \rightarrow \frac{z-1}{2}$ to get these roots

$$2 \left(\frac{z-1}{2} \right)^2 + 5 \left(\frac{z-1}{2} \right) + 10 = 0$$

$$\frac{2}{4} (z^2 - 2z + 1) + \frac{5}{2} (z-1) + 10 = 0$$

$$z^2 - 2z + 1 + 5z - 5 + 20 = 0$$

$$z^2 + 3z + 16 = 0$$

Alternatively

$$\begin{aligned} \text{Sum: } (2\alpha+1) + (2\beta+1) &= 2(\alpha+\beta) + 2 \\ &= 2 \times -\frac{5}{2} + 2 = -3 \end{aligned}$$

$$\begin{aligned} \text{Product: } (2\alpha+1)(2\beta+1) &= 4\alpha\beta + 2(\alpha+\beta) + 1 \\ &= 4 \times 5 + 2 \times -\frac{5}{2} + 1 \\ &= 16 \end{aligned}$$

$$\text{So } -\frac{b}{a} = -3, \frac{c}{a} = 16$$

$$\text{choose } a=1 \text{ then } b=3, c=16$$

$$\text{So } z^2 + 3z + 16 = 0$$

Advanced skills



EXAM PAPERS PRACTICE

Q5. $4z^2 + 3z - 1 = 0$, α, β roots

$$a) \quad \alpha + \beta = -\frac{b}{a} = -\frac{3}{4} \quad \alpha\beta = \frac{c}{a} = -\frac{1}{4}$$

new $2\alpha^{-1}, 2\beta^{-1}$

$$\text{Sum: } 2\alpha^{-1} + 2\beta^{-1} = \frac{2}{\alpha} + \frac{2}{\beta} = \frac{2\beta}{\alpha\beta} + \frac{2\alpha}{\alpha\beta}$$

$$= \frac{2(\alpha + \beta)}{\alpha\beta} = \frac{2 \times -\frac{3}{4}}{-\frac{1}{4}} = \frac{-6}{-1} = 6$$

$$\text{Product: } 2\alpha^{-1} \times 2\beta^{-1} = \frac{4}{\alpha\beta} = \frac{4}{-\frac{3}{4}} = -\frac{16}{3}$$

$$\text{so } 6 = -\frac{b}{a}, \quad -\frac{16}{3} = \frac{c}{a}$$

$$\text{let } a = 3 \text{ then } b = -18, c = -16$$

$$3z^2 - 18z - 16 = 0$$

$$b) \quad \frac{\alpha-1}{\beta}, \frac{\beta-1}{\alpha}$$

$$\text{Sum: } \frac{\alpha-1}{\beta} + \frac{\beta-1}{\alpha} = \frac{\alpha^2 - \alpha + \beta^2 - \beta}{\alpha\beta}$$

use $(\alpha+\beta)^2$ to get squared $\rightarrow$

$$= \frac{\alpha^2 + \beta^2 - (\alpha + \beta)}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta - 2(\alpha + \beta)}{\alpha\beta}$$

$$= \frac{\left(-\frac{3}{4}\right)^2 - 2 \times -\frac{1}{4} - 2 \times -\frac{3}{4}}{-\frac{1}{4}}$$

$$= \frac{\frac{9}{16} + \frac{1}{2} + \frac{3}{2}}{-\frac{1}{4}} = \frac{-\frac{9}{4} - 2 - 6}{-\frac{1}{4}}$$

$$= -\frac{41}{4}$$

Product: $\frac{\alpha-1}{\beta} \times \frac{\beta-1}{\alpha} = \frac{\alpha\beta - (\alpha+\beta) + 1}{\alpha\beta}$

$$= \frac{-\frac{1}{4} - -\frac{3}{4} + 1}{-\frac{1}{4}} = \frac{\frac{3}{2}}{-\frac{1}{4}} = -6$$

so $-\frac{b}{a} = -\frac{41}{4}$, $\frac{c}{a} = -6$

use $a = 4$ then $b = 41$, $c = -24$

so $4z^2 + 41z - 24 = 0$

Q6. $z^2 - kz + 6 = 0$ roots $\alpha, \alpha+1$

sum: $\alpha + \alpha + 1 = -\frac{b}{a} = k$ so $2\alpha + 1 = k$

Product: $\alpha(\alpha+1) = \frac{c}{a} = 6$

$$\alpha^2 + \alpha = 6$$

Solve simultaneous $\alpha = \frac{k-1}{2}$

$$\Rightarrow \left(\frac{k-1}{2}\right)^2 + \frac{k-1}{2} = 6$$

$$(k-1)^2 + 2(k-1) - 24 = 0$$

$$k^2 - 25 = 0 \quad k = \pm 5$$

Q7. $z^2 + 5z + 10 = 0$

$$\alpha + \beta = -\frac{b}{a} = -5, \quad \alpha\beta = \frac{c}{a} = 10$$

$$(\alpha + \beta)^3 = \alpha^3 + 3\alpha^2\beta + 3\alpha\beta^2 + \beta^3$$

$$\text{So } \alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha^2\beta - 3\alpha\beta^2$$

$$= (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$$

$$= (-5)^3 - 3 \times 10 \times -5$$

$$= -125 + 150 = 25$$

Q 8. $z^2 - 2z - 5 = 0$

$$\alpha + \beta = -\frac{b}{a} = 2, \quad \alpha\beta = \frac{c}{a} = -5$$

new $\alpha^2 + 1, \beta^2 + 1$

$$\begin{aligned}\text{Sum: } \alpha^2 + 1 + \beta^2 + 1 &= \alpha^2 + \beta^2 + 2 \\ &= (\alpha + \beta)^2 - 2\alpha\beta + 2 \\ &= 2^2 - 2(-5) + 2 \\ &= 16\end{aligned}$$

$$\begin{aligned}\text{Product: } (\alpha^2 + 1)(\beta^2 + 1) &= \alpha^2\beta^2 + \alpha^2 + \beta^2 + 1 \\ &= (\alpha\beta)^2 + (\alpha + \beta)^2 - 2\alpha\beta + 1 \\ &= (-5)^2 + 2^2 - 2(-5) + 1 = 40\end{aligned}$$

$$\text{So } -\frac{b}{a} = 16, \quad \frac{c}{a} = 40$$

so for $a = 1$ we have $b = -16, c = 40$

$$z^2 - 16z + 40 = 0$$